



IS216 Web Application Development II

Group Project
Briefing

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Sections: G3/G4



Synopsis

- To build a **dynamic and interactive web application** that utilizes the technologies you've learned this term including a back-end implementation for data storage and user authentication where necessary.
- The key learning outcome will be **demonstrating your ability to integrate** the various technologies you've studied and **apply them effectively in your final solution**.

Group Formation & Group Work

Step 1: Find your own group members



- Form your own group - choose people you can work and communicate with.
- Reach out early!!! There may be a group ready to adopt you. 🐱
- If you are still ungrouped near the deadline, speak to the teaching team.

Deadline: Week 4 lesson start time (register in [eLearn-G3/G4](#) → Groups)

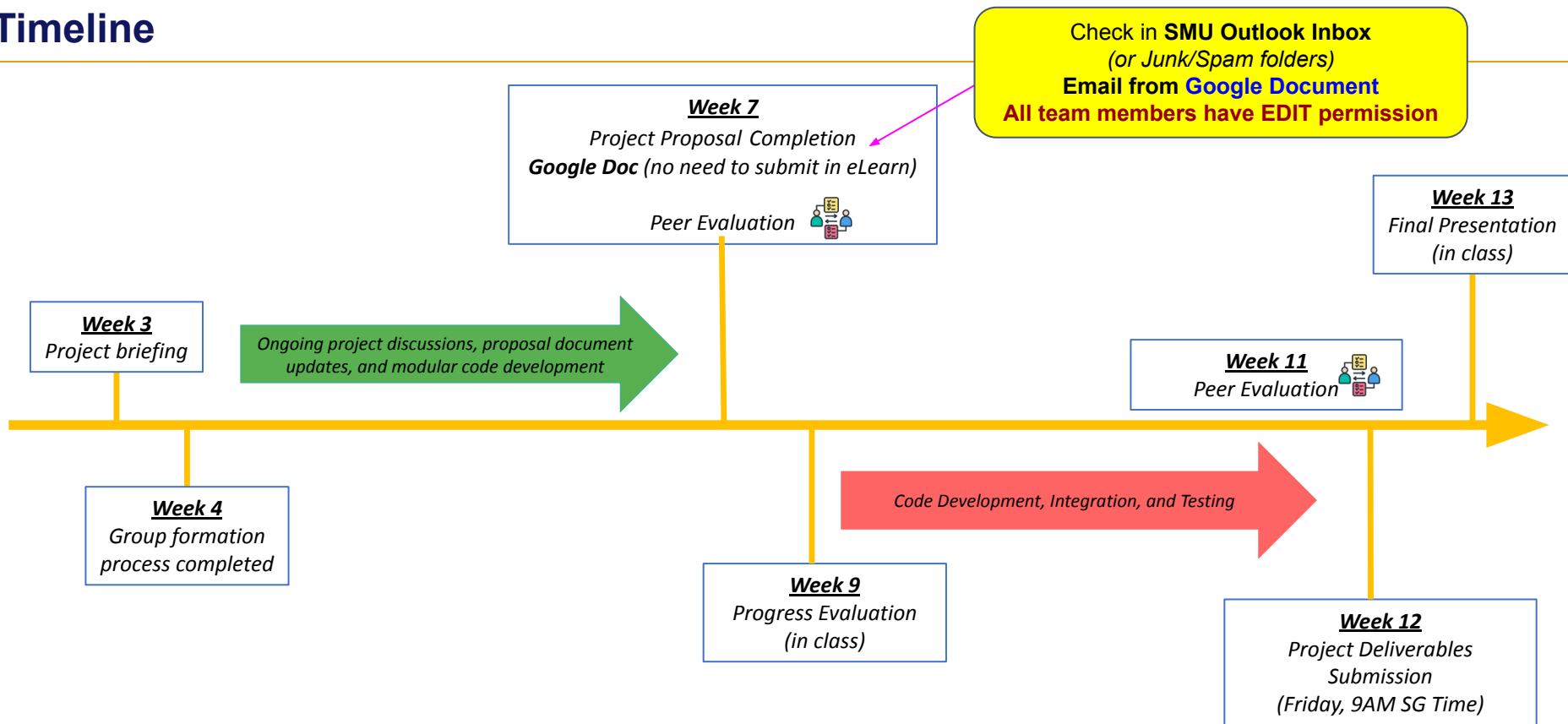
Step 2: Then, actually work as a group

- Agree on roles, expectations, communication channels, and milestones early.
- Share the workload and keep one another updated - no disappearing acts.
- Ask for help early if your group is stuck! Problems rarely improve by hiding.



Group Work, Explained in One Photo

Timeline



Timeline

- **Week 9** - Progress Evaluation (5% of the course grade)
 - 7 mins presentation + 5 mins Q&A
- **Week 12** - Friday 9AM SG Time
 - Submit **all project artefacts** (*see in the next slides for details*)

Submit within 1 hour of the specified deadline	10% penalty (of your score for the entire project)
Submit beyond 1 hour of the specified deadline	0 Mark

- **Week 13** - Final Presentation
 - 12 mins presentation (live demo of the web app) + 5 mins Q&A

Web Application :: Minimum Requirements

- You must **identify and solve a problem**.
 - *This can be either an unsolved problem or an existing problem.*
- You must use **HTML, CSS and JavaScript** in your implementation.
 - *Optionally, you are encouraged to use Bootstrap (or similar CSS framework), JSON, and Vue (or a similar JavaScript framework).*
- Your solution must include a **responsive design**.
- Your solution must be a **dynamic site** that **interact with a backend data store**.
 - *This could involve applying knowledge from IS113 (e.g. Express.js + MongoDB), but you are free to choose another data storage solution if you prefer (e.g. files, other types of databases, cloud storage).*
- Your solution must make meaningful use of **at least one external publicly available API** via asynchronous HTTP requests.
- Your solution must include **end-to-end (E2E) testing** for all core features.
- You must **utilize GitHub collaboration tools** available for your project to effectively manage version control, track changes, and facilitate communication among group members.
- You must **deploy your final solution to a remote server** (e.g. AWS, Google Cloud, Azure, Firebase, Vercel and etc.) that is accessible online.

Good Presentation

Try to incorporate what you've learned or you're learning in **Writing & Reasoning, Management Communication**, or **similar other module(s)**.

- Good **storytelling** (from the persona/user perspective)
- Do **NOT** read off slides, mobile phone, or notes
 - It's OK to prepare notes (bullet points) as a reference - you can peek into it once in a while but **DO NOT KEEP STARTING at your notes and read off it**
 - **PUT AWAY YOUR PHONE** (we will take your phone away - if they see you holding onto your phone)
- Eye gaze
 - Face and look at your audience, look around at your audience (not into the empty space)
- **Look** interested
- **Sound** interested
- Use **appropriate gestures**

Week 9 :: Progress Pitch (5%)

- Duration
 - 7 minutes pitch / group
 - 5 minutes Q&A (with the teaching team)
- You (as a group) must present (using slides / drawing if needed):
 - What is the **main business problem** you're trying to solve?
 - Who is the **main audience** of the proposed web application? (*one or more personas*)
 - What is the **main objective** of the proposed web application?
 - How does it help business/people/society?
 - How does it help remove/relieve the main audience's pain points or fill any gaps in their user journey?
 - Describe the web application.
 - What are the **main components/features** of the app?
 - What will the main audience **experience**?
 - Describe this **user journey** end-to-end in a storytelling manner.
 - Group members, their roles, and **technical responsibilities**
- ALL members **must be present** during the in-class/F2F pitch.
 - Missing members (or late arrivals) will be penalized with ZERO (0) mark for the proposal grade.
 - All members must be ready to answer questions from the teaching team and provide input.
- What **design/development work have you done so far**?
 - Wireframes (e.g. FIGMA)
 - Setting up a group project GitHub repository
 - Infrastructure (e.g. cloud set up)
 - Any feasibility testing (e.g. 3rd party APIs)
 - Coding

Week 9 :: Progress Pitch (5%) :: Grading Rubrics

Component	Description	%
Problem Definition	Clearly articulates the problem being addressed, including goals and constraints.	25%
Concept Design & Technology Stack	Outlines the proposed solution, including design approach and chosen web technologies/frameworks.	25%
Task Delegation	Demonstrates effective group collaboration through clear distribution of roles and responsibilities.	25%
Feature Demonstration	Live demo of at least one working feature showcasing functionality and progress. <i>e.g. API call, CRUD operation, feature or sub-feature</i>	25%
		100%

Week 9 :: Progress Pitch (5%) :: Grading Rubrics

Marks	Details
C+ & below	Problems / features / job scopes / web design / tech stacks are vague. There is no functioning task.
B- to B+	Problems / features / job scopes / web design / tech stacks are not very clear but somewhat defined. A task is somewhat functioning.
A- to A	Problems / features / job scopes / web design / tech stacks are clearly defined. A task is functioning well.
A+	Challenging Problems / features / job scopes / web design / tech stacks are clearly defined. A challenging task is functioning well.

Week 12 :: Final Deliverables & Submission in eLearn (G3/G4)

- By the end of Week 12 Friday 9AM SG Time, please submit your group's **deliverables** in [eLearn \(G3/G4\)](#) → **Assignments** → **[Project] Final Deliverables**
- What do you need to submit in **eLearn**?
 - 1) Record your web application video (maximum 12 minutes, minimum 720p - ideally 1080p), upload to **YouTube**
 - You can make **either** of the 2 video types:
 - **Full presentation**
 - This will be an online equivalent of your typical class presentation (except, we do not need to see your faces).
 - **Marketing pitch**
 - This will be like those short/quick marketing advertisement videos you see in YouTube - and it will be shorter < 2 mins.
 - Examples: [Grammarly](#), [Monday.com](#), [Moodcaster](#)
 - Marketing pitch videos do NOT go into the details of the inner-workings of the system. Rather, it purely focuses on the **values added from the user's perspective**.
 - 2) **Presentation Slides** (PPTX or PDF)
 - A link/URL to your **video (presentation or pitch)** should be available on **Slide #1**.
 - Put the **link/URL** to your **deployed (to cloud) web app** on **Slide #1**.
 - 3) **Source Code**, complete **BOTH of the following**
 - **(Step 1)** Zip up your web application (including ALL files) and submit the ZIP file (***note eLearn has a max file size limit***)
 - If your **zip file is too large**, upload it to **OneDrive folder**, make it **SMU-viewable**, put the link to the folder on an appendix slide in your **presentation slides**.
 - **(Step 2)** Push your web application (including ALL files) into **GitHub** and **include the LINK** to your **repo** in your **Presentation Slides (appendix slide)**
 - You can make it **public** or **private** (if make private, send invite to **Kyong** at kjshim@smu.edu.sg and **Chein** at ccwong@smu.edu.sg).

Week 13 :: Final Presentation & Web Application (20%)

- Venue: **SCIS1 Computing Lab B1-1**
- Duration (total maximum 17 mins - strictly applied)
 - 12 minutes presentation / group
 - 5 minutes Q&A + feedback (*with the audience*)
- You (as a group) must present (using slides / demo):
 - **Introduce your group in < 20 seconds** (“Hi, we are Group X - group name”) and **MOVE ON**
 - Background (max 2 mins)
 - What is the **main business problem** you’re trying to solve with your proposed **web application**?
 - Who will **benefit** from your solution and how?
 - Demo your group’s **web application** (you should spend most of your presentation time on this)
 - Your demo must be showing a **live/working web application**
 - **NO playing of pre-recorded videos is permitted**
 - Instead of describe your web app, do **storytelling** from the **main user’s perspective**
 - You do NOT need to show EVERYTHING about your web application (**YOU HAVE LIMITED TIME !!!**)
 - Pick and choose - show **most interesting stuff** that aligns with your **storytelling**.
- ALL members **must attend and present** (**stand in front of the classroom together & present their respective part**).
 - Missing members, late arrivals, and non-presenting-members will receive ZERO (0) mark for the group project.
 - All members must be ready to answer questions from the audience.

We have a **very tight schedule** in **Week 13**.
NO extra time will be granted whatsoever.

If possible, prepare a **backup computer** (that you can switch to right away in case the **primary computer** cannot be used for **web app demo**). It is **YOUR (students’) responsibility to ensure this**.

Week 13 :: Final Presentation & Web Application (20%) :: Grading Rubrics

Component		%
(1) Presentation		25%
	The presentation	20%
	Q & A	5%
(2) Working web application		75%
	Problem(s) solving and solution appropriateness	20%
	Working app, correctly implemented and usable solution	27%
	Styling (including responsive design)	18%
	Testing	10%
		100%

Week 13 :: Final Presentation & Web Application (20%) :: Grading Rubrics

Presentation

Marks	Details
C+ & below	Presentation with Demo is sloppy. Difficult to understand how the solution addresses the problem, or the clarity of the technologies used.
B- to B+	Presentation with Demo is of reasonable standard. Relatively clear presentation and demo that shows how technologies are used to address the problem with some minor hiccups.
A- to A	Presentation with Demo is good. Clear presentation and demo that are quite seamless. Easy to understand, with a clear explanation of what was done and how it was accomplished.
A+	Presentation with Demo is exceptional. Crystal-clear presentation and demo that are seamless. No clarification questions are needed to understand what was done. Additionally, at least one bonus point must be earned to qualify.

Week 13 :: Final Presentation & Web Application (20%) :: Grading Rubrics

Web Application

- Problem(s) solving and solution appropriateness

Marks	Details
C+ & below	Problem is not solved and/or not a useful app.
B- to B+	Problem is somewhat solved and is relatively useful. The problem might not be completely solved but is mostly addressed.
A- to A	Problem solved is very useful. A clearly defined problem that is addressed with a satisfactory solution.
A+	Challenging problem is solved and very useful. It is a challenging problem to address, and the solution is highly appropriate.

Week 13 :: Final Presentation & Web Application (20%) :: Grading Rubrics

Web Application

- Working app, correctly implemented, usable solution*

Marks	Details
C+ & below	App is somewhat working to not working. Intermittently working, with some parts functioning or completely non-functional.
B- to B+	App is buggy with some working parts. Contains bugs, but none are critical or show-stopping. The app is roughly usable without requiring excessive effort to understand how to use it.
A- to A	App is working well, and usability is good. Some minor bugs, but the app is relatively intuitive to use.
A+	App is working perfectly with great usability. No bugs at all; the app is completely efficient and intuitive, requiring no effort to figure out how to use it.

*In the sense of the following guidelines:

<https://www.interaction-design.org/literature/article/user-interface-design-guidelines-10-rules-of-thumb>

Week 13 :: Final Presentation & Web Application (20%) :: Grading Rubrics

Web Application

- Styling (including responsiveness)

Marks	Details
C+ & below	Limited styling implemented; Minimal responsiveness implemented.
B- to B+	Relatively decent styling and somewhat responsive site Somewhat clean and appealing, and the site is somewhat responsive, with minor issues but nothing critical.
A- to A	Good style and somewhat responsive site Attractive and well-executed, and the site is fully responsive.
A+	Excellent style and fully responsive site A beautiful site that makes effective use of style and is fully responsive, scaling appropriately for different screen sizes and orientations.

Week 13 :: Final Presentation & Web Application (20%) :: Grading Rubrics

Web Application

- Testing

Marks	Details
C+ & below	Limited coverage of core features. Tests are limited, generally non-repeatable and not stable. Provide very limited instructions in README on how to run tests.
B- to B+	Decent coverage of core features Tests cover some of the app's main user journeys. Some of the tests are repeatable and stable. Provide some brief instructions in README on how to run tests. Tests demonstrate somewhat understanding of testing concepts.
A- to A	Good coverage of core features with some quality test cases. Tests cover most of the app's main user journeys. Tests are mostly repeatable, generally stable and have clear structure. Provide somewhat clear instructions in README on how to run tests. Tests demonstrate a good understanding of testing concepts.
A+	Excellent coverage of core features with quality test cases. Tests cover the app's main user journeys. Tests are repeatable and use stable selectors and have clear test structure. Provide clear instructions in README on how to run tests. Tests demonstrate excellent understanding of testing concepts.

Week 7 and 11 :: Peer Evaluation (5%)

- Peer Evaluation Submission:
 - Every team member must **submit 2 times of peer evaluation form (week 7 and week 11)**.
- Internal Conflict Resolution:
 - **Teams must actively manage and resolve conflicts internally** instead of waiting to report them on the peer evaluation form.
- Early Intervention:
 - **Report unmanageable team members to the teaching team by week 7.**
- Shared Responsibility:
 - If a team fails to attempt conflict resolution, all members may receive the same grade regardless of peer scores.
- Grading Criteria:
 - Teamwork and project management are implicit components of the final project grade.
- Severe Penalties:
 - **Members who fail to contribute significantly or miss most meetings face strict penalties**, up to a zero mark for the entire project.

Week 13 :: Self & Peer Evaluation in eLearn (G3/G4)

- Please complete [Group Project] End-Term Self and Peer Evaluation in [eLearn \(G3/G4\)](#)
- Deadline: 15-NOV-2026 (Sunday) 11:59 PM SG Time (*end of Week 13*)
- All students are required to complete this evaluation.
- By default, all members of the group receive the same project grade.
 - However, if you receive poor peer evaluation scores, the teaching team will assign you a lower grade after an investigation.

Action Items

- **Appoint a leader** for your group to guide and manage the project effectively
- Your group should **begin meeting, brainstorming, and discussing the project**
- Document all your discussions in the **project proposal**
 - *TA will be sending out the template to each group (Google Document)*
- Your group should finalize the project idea and conduct a final review of the project proposal document before the **Week 7 Friday 11:59 PM SG Time**

Common Project Ideas - **Consider Something Different !!!**

- Event Management Portal
- Donation Drive Portal
- Recipe/Meals Finder Portal
- Job Finder Portal
- Matchmaking Portal
- Pet Adoption Portal
- Volunteer Recruitment Portal

**Don't start with
"What portal can we build?"**

**Start with
"What interesting problem can we solve?"**

You may still propose these topics, but ***we strongly encourage you to explore a less common problem domain or bring a genuinely new perspective to an existing one.***

Frequently Asked Questions (FAQ)

- Are we required to use **Bootstrap** or **Vue.js**?
 - No, you are not required to use Bootstrap or Vue.js specifically. However, you are encouraged to use frameworks to **simplify, standardize, and speed up** the development process of your project.
- I want to use React. Is this allowed?
 - Yes, using React or a similar framework is **allowed**. However, we strongly **encourage you to use Vue.js** for the following reasons:
 - **Technical Support**: Vue.js is covered in the syllabus, so we can provide more effective support. React is not officially included in the course materials.
 - **Exam Preparation**: The final exam will include questions on JavaScript and Vue.js, but not on React. Using Vue.js will better prepare you for the assessment.
 - Also, consider your **group mates' familiarity with React**. If everyone is comfortable with it, using React is fine. However, if some group members are not, they would need to learn two different JavaScript frameworks during the course. Make sure the whole group is comfortable with this decision.

Frequently Asked Questions (FAQ)

- Can the teaching team provide technical support on the technology we are using, e.g., Docker, AWS, etc.?
 - **In general, no.** There is a plethora of technologies out there. Given the variability in our expertise with different technologies, it would be unfair for us to assist one group with technology we are familiar with while being unable to support another group with technology, we are not knowledgeable about. **However, we are happy to engage in high-level discussions regarding the appropriateness of different technologies for your solution.**
- Which browser should we use?
 - You can **use any browser you prefer.** However, for grading purposes, please ensure that your website displays properly in Google Chrome. **We will also use Chrome's developer tools to test your site's responsive design.**

Frequently Asked Questions (FAQ)

- Are we allowed to use publicly available code sources (e.g. from GitHub projects)?
 - You **may use publicly and freely available libraries** (i.e., not paid). However, such usage must be clearly identified in your presentation and within the comments of your code. You **must NOT present this code as your own work**, as that would constitute plagiarism. If you are unsure, please contact your teaching team to discuss your plans.
- Are all members of a group receiving the same group project grade?
 - Yes, a single grade is awarded for the project. However, if a group member is not participating, please inform your instructors **EARLY**, and individual adjustments will be considered on a case-by-case basis. However, if you notify your instructors too close to the final deliverable deadline (e.g., one week before), it may be considered too late to take any meaningful action, and your instructors may not accept your case.

Frequently Asked Questions (FAQ)

- Are all members of a group required to code?
 - Yes, all members should contribute to the coding of the project. It is advisable to assign specific tasks to each member (e.g., Bob handles styling, and Olivia manages scripting) to ensure effective collaboration.
- Are all members of a group required to present?
 - Yes, all members should take this opportunity to practice their presentation skills.
- Are we allowed to use back-end code from IS113 and build on it?
 - Yes, you are allowed to build upon example back-end code from IS113. In IS216, we will provide an example of how to apply styling and scripting to some back-end code, and you are welcome to build on that as well if you wish.

Frequently Asked Questions (FAQ)

- Will you be grading the backend code?
 - While there is no explicit grading component specifically for the back-end code, the overall functionality of your application is evaluated. Therefore, the quality of your back-end implementation will be assessed under the grading criterion for a “working app, correctly implemented, and usable solution.”
- If other groups are tackling the same problem, would we be penalized?
 - No. There are often many ways to solve the same problem, and multiple groups working on similar issues is acceptable.

Frequently Asked Questions (FAQ)

- Can I use dummy data instead of real-data?
 - If the type of data your application requires is publicly available, we encourage you to use real data to make your prototype more realistic. However, if you anticipate difficulties in obtaining real data, please consult with your teaching team for guidance.
- Under “problem solving and solution appropriateness”, are you referring to coding complexity, or more to the business side?
 - We are referring to the business side, meaning you should clearly articulate the problem your website addresses and demonstrate that the implementation effectively solves this problem. The problem does not need to be a critical societal issue. For example, a virtual escape room may not address a major societal problem but still solves a relevant issue, such as enabling remote connection with friends and family, especially during a pandemic.

Frequently Asked Questions (FAQ)

- Are we allowed to scrape data from a website?
 - If you choose to scrape data from a website, you must ensure that you comply with any restrictions outlined in the site's robots.txt file and consider any legal implications. Additionally, you must clearly identify and properly credit the sources of your data. Note that using scraped data does not substitute the requirement to utilize at least one publicly available API.
- How will the responsive design of our web application be tested?
 - We will test your website using Chrome's developer tools across various viewports, starting from the iPhone 6 to the XL breakpoint of Bootstrap. Ensure that your layout looks good and adjusts properly across all viewports within this range.

Frequently Asked Questions (FAQ)

- The API I'm using is free, but only up to certain amounts of usage. How much testing will you be doing after submission?
 - As a rough estimate, each functionality will be tested up to 5-6 times per day. You will need to assess whether this falls within the free usage limits of your API.

Frequently Asked Questions (FAQ)

- Are we allowed to use AI/LLM in the project?
 - Generally, we allow the use of AI/LLM for lower order thinking tasks such as:
 - information search
 - Generating website concepts, layouts, or themes
 - Exploring different UI/UX design inspirations
 - Explaining coding errors and debugging hints
 - Boilerplate code generation – starter code and code snippets
 - Generating unit test cases, sample inputs, or mock data
 - You are not allowed to use AI/LLM for higher order thinking tasks such as:
 - Core implementation tasks (to implement main business logic, backend endpoints, or critical frontend interactivity)
 - Problem solving – students must demonstrate their own capability for solving problems and addressing major implementation issues, not just rely on automated AI fixes.
 - In the README.txt file, state clearly the parts you solely rely on AI/LLM.